Social Information Filtering-Based Electricity Retail Plan Recommender System for Smart Grid End Users

Fengji Luo, *Member, IEEE*, Gianluca Ranzi, *Member, IEEE*, Xibin Wang, and Zhao Yang Dong, *Fellow, IEEE*

***Abstract*—Rapid growth of data in smart grids provides great potentials for the utility to discover knowledge of demand side and design proper demand side management schemes to optimize the grid operation. The overloaded data also impose challenges on the data analytics and decision making. This paper introduces the service computing technique into the smart grid, and proposes a personalized electricity retail plan recommender system for residential users. The proposed personalized recommender system (PRS) is based on the collaborative filtering technique. The energy consumption data of users are firstly collected from the smart meter, and then key energy consumption features of the users are extracted and stored into a user knowledge database (UKD), together with the information of their chosen electricity retail plans. For a target user, the recommender system analyzes his/her energy consumption pattern, find users having similar energy consumption patterns with him/her from the UKD, and then recommend most suitable pricing plan to the target user. Experiments are conducted based on actual smart meter data and retail plan data to verify the effectiveness of the proposed PRS.**

***Index Terms*—Smart grid, service computing, recommender system, demand side management, energy management system.**

I. Introduction

VARIOUS wide-area and distributed data sources in today's power grids generate unprecedented data volumes in the grids' daily operations. For example, a provincial utility grid in China can reach 20 million meters. And if one considers the data is collected every 5-15 minutes, this grid alone will generate about 1-terabyte a day [1]. The large available data sets provide opportunities for the utility to learn knowledge from the data and make proper strategies to optimize the grid operation, but also impose significant challenges on the data collection, transmission, and analytics [2], [3]. Meanwhile, driven by the advances in Web technology, there is an ongoing trend for transforming utility services to be an E-commerce. For example, the Public Utility of Commission of Texas, the U.S., has established the "Power-to-Choose" Web platform for selling the electricity retail plans to the end users [4]. The end users can view, compare, and choose the retail plans from different retailers on the website.

Above development trends provide new opportunities for the utility to learn end users' needs and preferences on the electrical services/products, and design efficient demand side energy management systems (EMSs) to maximize the users' response rates and improve the grid energy efficiency. Currently, the prevalent demand side management (DSM) techniques can be generally categorized into two classes: Incentive based DSM programs and pricing based DSM programs.

A. Overview of Demand Side Management Techniques

The incentive based DSM technique is also known as direct load control (DLC) [5]. In DLC, the utility directly controls the settings or ON/OFF status of electrical resources of the end users through remote controllers and actuators. These electrical resources could include household appliances, electric vehicles (EVs), etc. The utility would provide incentives or rewards to the users as the subsidy of the load control. DLC control techniques has been widely studied in the literature. For example, many works study the direct control of the thermostatically controlled loads (TCLs) to support the grid-level applications [6], [7]. Some researchers investigate the coordinated direct control scheme of various household appliances by considering the users' utilization preferences [8]. There are also some DLC industrial projects implemented around the world, such as the 'SmartAC' program of Pacific Gas & Electric (PG&E) [9].

The pricing based DSM technique is also known as indirect load control [10]. In the pricing based DSM programs, the utility sets up a certain pricing signals to stimulate users actively

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F. Luo and G. Ranzi are with the School of Civil Engineering, University of Sydney, NSW 2006, Australia (e-mail: fengji.luo@sydney.edu.au; gianluca.ranzi@sydney.edu.au).

X. Wang is with the School of Electrical and Information Engineering, Guizhou Institute of Technology, Guiyang 550003, China (e-mail: binxi.wang@cqu.edu.cn).

Z. Y. Dong is with the School of Electrical Engineering and Telecommunications, University of New South Wales, NSW, Australia (e-mail: zydong@ieee.org).

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adjust their appliance utilization time to the periods with lower rates. The widely adopted variable pricing schemes mainly include the Time-of-Use (TOU) pricing [11], real-time pricing (RTP) [12], and critical peak pricing (CPP) [13], refers to the programs in which customers respond to the time-varying changes in retail electricity tariffs. TOU rates can achieve greater benefits, but its main drawback is that its structure and rates are not frequently updated. RTP is a dynamic pricing scheme with a price updating cycle of one hour or less. RTP can more accurately reflect the demand-supply balance and marginal cost variations in a power grid, but its frequent update cycles are often too complex for residential customers. CPP is a compromise between TOU and RTP. In a CPP program, TOU rates are still applied in normal periods. When a peak event is predicted by the control centre, it will send a notice to customers to indicate that a critical peak period, in which a much higher critical peak rate will be applied.

B. A New Type of Energy Management System: Smart Grid Personalized Recommender Systems

According to [14], demand response (DR) programs can be implemented at different levels: manual DR, semi-automated DR, and fully-automated DR. Existing residential energy management systems (REMSs) often work in fully-automated and semi-automated DR environments, in which energy management systems directly schedule and control the residential energy resources (energy storage systems, appliances, plug-in electric vehicles, etc.). Currently, the manual DR is often based on the aforementioned varying pricing signals. Our recent researches [15], [16] show that even in the manual DR environment, there is still the opportunity to develop a typology of REMS to enhance the household energy efficiency. That is, to introduce the personalized recommendation (PR) (also known as service recommendation or recommendation) technique to capture the user's needs and interests, and recommend energy-aware products/suggestions to the user.

With the prevalence of Web technology and data explosion, PR technology has attracted wide attentions in both the computer science academia and industry [17], [18]. PR provide powerful tools to learn users' preferences and interests from a large set of available user-related data (e.g., bank transaction records, user comments, etc.), and recommend personalized services/products to users in which they are mostly interested. In the last a few years, recommendation techniques have already been widely applied in the e-commerce industry, and many personalized recommender systems (PRSs) have been developed to recommend different kinds of products to the users, such as movies, books, Web services, etc.

As mentioned before, modern power systems are experiencing a trend to deliver Web based electronic services to end users. Therefore, there could be huge potentials for the utility to apply PR technique to obtain end users' needs and interests from the residential energy consumption data, and recommend different kinds of energy-aware products/services to the users. In [15], we discussed the vision of applying PR technique into smart grid, and proposed conceptual designs of three applications. Those application scenarios represent a new type of REMS, which is referred as the smart grid personalized recommender system (SGPRS). SGPRS is essentially a kind of data-driven, networked energy management system. It does not rely on incentive nor pricing signals to optimize the user's energy consumption, but implicitly find out mostly suitable, energy-aware products/services for a specific user from a large number of available items. In [16], we implemented one of the three applications in [15], which is an energy saving appliance recommendation system based on the information retrieving and non-intrusive load monitoring techniques.

C. Scope of This Paper

As aforementioned, developed countries have been transforming the power retail business to be an E-commerce. Also in the developing countries, deliberation of the power retail market is an ongoing trend [19]. For example, recently China has made strategic plans to open the electricity retail market, and established trial retail markets in some cities, in which a large number of retailers are authorized [20]. In this paper, following our former works [15], [16], we introduce PR into DSM and develop another specific application of SGPRS—a *personalized electricity retail plan recommender system*, which is the second application discussed in [15]. In this application, the utility analyzes the energy consumption features of a user, and recommend mostly suitable electricity retail plans to the user, so as to improve the energy efficiency of the demand side. The proposed PRS is based on the collaborative filtering (CF) recommendation technique, which filters the items from the social information of a group of users. For a target user, the proposed PRS extracts his/her household energy consumption features, and recommends suitable electricity retail plans to the user by analyzing the choices of a group of users who have the similar energy consumption features with the target user.

This paper is organized as follows. Section II gives an overview on the personalized recommendation techniques; Section III introduces the settings of typical electric retail plans; Section IV presents the proposed personalized electricity retail plan recommender system; Section V discusses the experiment results; and Section VI gives conclusions and future works.

II. Introduction of Personalized Recommendation

With the overloaded information of today's society, people often need to make choices on their interested or needed products/services from a large number of available ones. Driven by this, PR techniques are developed to automatically learn users' preferences and interests, and extract and recommend meaningful information to users. PR technique has gained large successes in the field of E-commerce. For example, as the first E-commerce website which applies the PR technique on its business, Amazon has a successful implementation on the precision marketing. Statistical data show that 35% of Amazon's sales are provided by its product recommendation engine [21].

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To make readers more familiar with the research in this paper, in this section we firstly give a brief overview of PR techniques, and then particularly introduce the basics of the CF recommendation technique, which is applied in the design of the proposed PRS in this paper.

A. Overview of Personalized Recommendation

In despite of their prevalence in both computing academia and industry in recent years, the proposal of personalized recommendation techniques actually can be traced back to the mid-1990s [24]. From then, many PRSs have been developed, aiming to assist people to deal with information overload and provide personalized item recommendations to them. For example, in our recent works, we developed a support vector machine (SVM) based electronic movie recommendation system [22] and a quality-of-service (QoS) aware Web service recommendation system [23].

Formally, a recommendation problem can be described as follows [17]. Let C and S be the sets of users and possible items which can be recommended, respectively. And let u be a utility function that measures the usefulness of item s to user c, i.e., $u : C \times S \rightarrow R$, where R is a totally ordered set of nonnegative integers of real numbers. Then, the recommender system chooses the items $s'_c \in S$ for a given user $c \in C$, to maximize the user's utility,

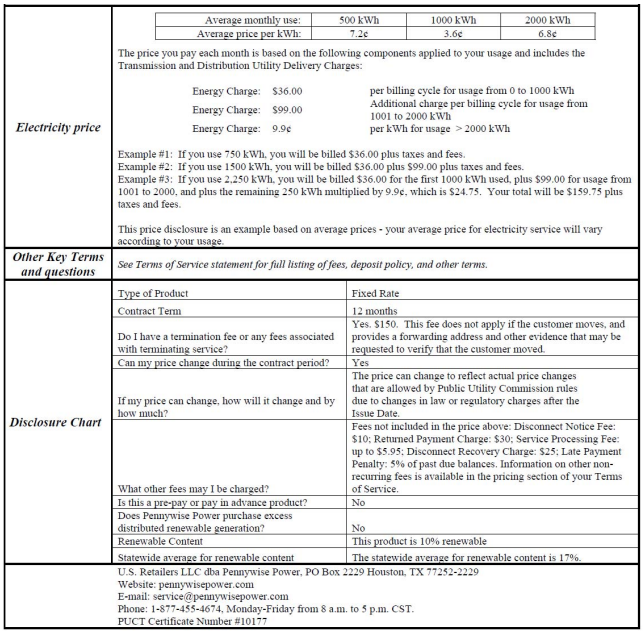
$$s'_c = \arg\max_{s \in S} u(c, s) \tag{1}$$

In personalized recommendation, the utility of an item is usually represented by a rating of a user, which can be a binary value to indicate 'like' or 'dislike', or numerical value to indicate the preference degree of a user on a particular item [17]. Note here the meaning of the word 'utility' is different with the power utility. In most situations, the utility u is often not defined on the whole $C \times S$ space, but only a subset of it. That is, users only rate a subset of items [18]. Therefore, the core task of PR is to predict unknown ratings of the user on nonrated items from the known ratings, and make appropriate recommendations to the user based on those predictions.

The most commonly used recommendation methods can be categories into two classes: content-based recommendation methods and collaborative filtering based recommendation methods [25]. Content-based recommendations recommend items to a user based on the similar items the user preferred in the past. For example, a book recommendation system will learn the commonalities among the books which have been highly rated by a specific user in the past. Then the books which have a high degree of similarity to whatever the user's preferences will be selected and recommended to the user. Unlike content-based recommendations, CF based recommendations try to predict the score of items for a particular user based on the items previously rated by other users. It finds the "peers" of a specific user who have similar tastes. Then, only the items that are most preferred by the "peers" will be recommended [17], [18].

B. CF Based Recommendation Technique

CF based recommendations aim to make rating predictions for a user on a nonrated item, based on the similar users.



Electricity price	Average monthly use: 500 kWh 1000 kWh 2000 kWh Average price per kWh: 7.2¢ 3.6¢ 6.8¢ The price you pay each month is based on the following components applied to your usage and includes the Transmission and Distribution Utility Delivery Charges: Energy Charge: $36.00 per billing cycle for usage from 0 to 1000 kWh Energy Charge: $99.00 Additional charge per billing cycle for usage from 1001 to 2000 kWh Energy Charge: 9.9¢ per kWh for usage > 2000 kWh Example #1: If you use 750 kWh, you will be billed $36.00 plus taxes and fees. Example #2: If you use 1500 kWh, you will be billed $36.00 plus $99.00 plus taxes and fees. Example #3: If you use 2,250 kWh, you will be billed $36.00 for the first 1000 kWh used, plus $99.00 for usage from 1001 to 2000, and plus the remaining 250 kWh multiplied by 9.9¢, which is $24.75. Your total will be $159.75 plus taxes and fees. This price disclosure is an example based on average prices - your average price for electricity service will vary according to your usage.
Other Key Terms and questions	*See Terms of Service statement for full listing of fees, deposit policy, and other terms.*
Disclosure Chart	Type of Product: Fixed Rate Contract Term: 12 months Do I have a termination fee or any fees associated with terminating service?: Yes. $150. This fee does not apply if the customer moves, and provides a forwarding address and other evidence that may be requested to verify that the customer moved. Can my price change during the contract period?: Yes If my price can change, how will it change and by how much?: The price can change to reflect actual price changes that are allowed by Public Utility Commission rules due to changes in law or regulatory charges after the Issue Date. What other fees may I be charged?: Fees not included in the price above: Disconnect Notice Fee: $10; Returned Payment Charge: $30; Service Processing Fee: up to $5.95; Disconnect Recovery Charge: $25; Late Payment Penalty: 5% of past due balances. Information on other non-recurring fees is available in the pricing section of your Terms of Service. Is this a pre-pay or pay in advance product?: No Does Pennywise Power purchase excess distributed renewable generation?: No Renewable Content: This product is 10% renewable Statewide average for renewable content: The statewide average for renewable content is 17%.
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Fig. 1. Sample of the electricity retail plan [4].

Currently, the collaborative based recommendation techniques can be generally classified as two kinds: memory-based techniques and model-based techniques [26].

Memory-based techniques generally make rating prediction from the entire collection of items which are previously rated by the users [27]. The predicted rating of a user c on a nonrated item s, $r_{c,s}$, is computed as the aggregate of the ratings from the similar users:

$$r_{c,s} = \underset{c' \in C^*}{agg}\,(r_{c',s}) \tag{2}$$

where C^* is the set of similar users of user c and $agg(\bullet)$ is the aggregation function. Different aggregation functions can be used, such as the average function or weighted average function. Some similarity calculation functions can be applied to calculate the similar degree of two given users, such as the cosine function and Pearson correlation [18].

Model-based techniques utilize the previously ratings to train a model, and then use the model to predict the rating of a user on a nonrated item. Different approaches can be used to do the model training, such as the probabilistic approach [26], Bayesian networks [28], maximum entropy model [29], etc.

III. Configuration of Electricity Retail Plan

The choices of end users on electricity retail plans are heavily determined by the plan configurations. In this study, we consider the electricity retail plan configurations from the Power-to-Choose website [4]. Currently there are more than 1,000 retail plans from different electricity retail companies of Texas are published on the Power-to-Choose platform. Users can login the website by inputting their post-codes, and then check the detailed information of different plans. In this section, we give a general introduction on the configuration of the retail plans in the Power-to-Choose platform, which we will be used to produce our simulation data later. The introduction

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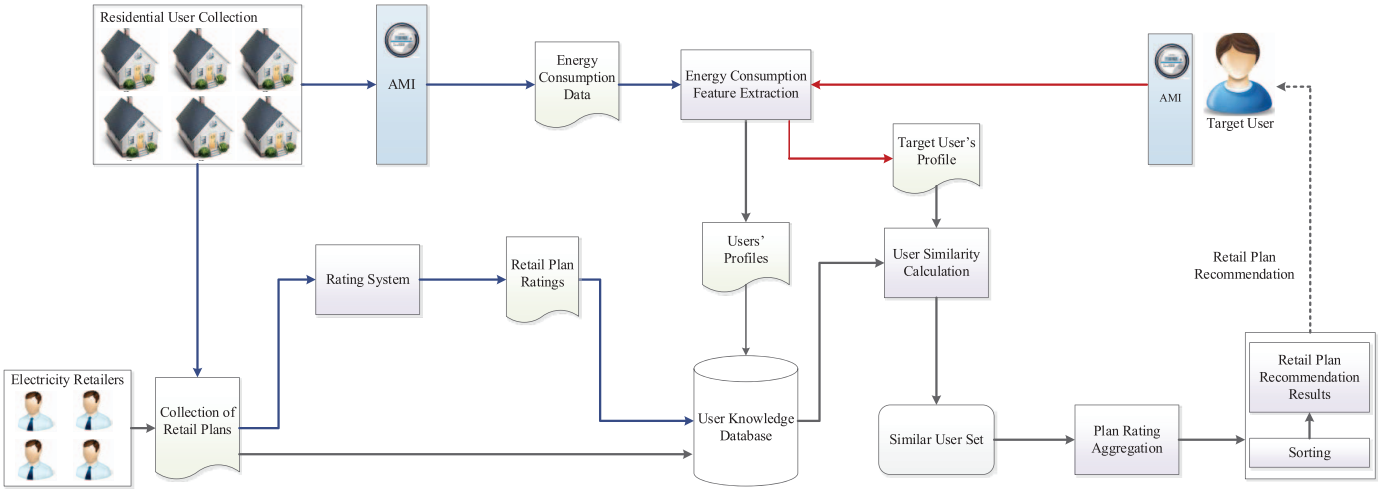


Fig. 2. Schematic of the personalized electricity retail plan recommender system.

we give in this section can be also used as a useful reference for readers who are interested in the Texas electricity retail market.

Fig. 1 shows a sample of the electricity retail plan in the Power-to-Choose platform. The main configurations of the plan often consist of following four parts:

(1) Transmission and distribution utility (TDU) charges. Currently, the transmission and distribution services of Texas are undertaken by six transmission/distribution companies, and the retail companies collect the TDU charges on behalf of them. The TDU charges are calculated as:

$$TDU = R_{fix} + R_{unit} \times E_m \tag{3}$$

where R_{fix} is the monthly fixed rate ($); R_{unit} is the unit rate ($/kWh); and E_m is the monthly energy consumption of the user (kWh). In some plans, the TDU charges are integrated into the energy charges for calculation convenience purpose;

(2) System administration fee. This fee is charged by the Electric Reliability Council of Texas (ERCOT), and is fixed as 0.0555$/kWh;

(3) Energy charge. This part is the main body of the retail pan. There are two kinds of energy charge rates: time-of-use (TOU) rate and non-TOU rate. For the non-TOU plan, a fixed rate is used. The TOU plan often applies different rates in working periods and non-working periods (evenings and weekends). For both non-TOU and TOU plans, there are further different rates applied: (a) *fixed rate*, which is fixed during the whole contract period; (b) *variable rate*, which is variable in the monthly basis and determined by the retail company; (c) *index rate*, which is also variable along with a public index. Some fixed rate, non-TOU plans adopt the progressive charging scheme, where the rate per kWh increases or decreases with the increase of monthly energy usages;

(4) Base charge and minimum usage charge. Base charge is a monthly fixed fee. The minimum usage charge is an extra charge in case the user's energy consumption in a month is less than the contracted minimum consumption threshold. The values of both charges are often in the range of [$5, $15].

Each plan is also associated with a contract length. A certain cancelling fee might be charged if the user terminates the contract before its expiration.

IV. Design of the Electricity Retail Plan Recommender System

In this paper, a PRS is developed to recommend suitable electricity retail plans to end users, based on their energy consumption analysis. The proposed PRS exploits the tastes of different users on retail plan, and recommend plans to a target user. Firstly, the working of the PRS is based on following assumption.

Assumption: There are trends or patterns on choosing the electricity retail plans between groups of people.

With the continuous increase of end users' awareness on demand response, above assumption is reasonable in modern grid to a large extent. For example, a user who often has large energy consumption deviations during the different periods of a day would prefer the TOU plan. Another example is that the users would prefer the plans published by retailers with high reputations. Therefore, the aiming of the proposed PRS is to discover such trends and patterns from the collective preferences of the user groups, and then make recommendations to the target user.

For a target user, the PRS finds out his/her similar users from a database by calculating the user similarities, and then it aggregates choices of the similar users on retail plans to form a recommendation list. The schematic of the proposed PRS is shown as Fig. 2, and its working mechanism is summarized as below.

(1) The PRS establishes a *user knowledge database (UKD)*. The UKD stores prior knowledge of a set of users, including: (a) users' energy consumption features, which are extracted from their energy consumption data collected by the Advanced Metering Infrastructure (AMI); and (b) the information of electricity retail plans historically chosen by the users;

(2) The PRS uses a rating inference system to implicitly produce the ratings of the users on their current & historically

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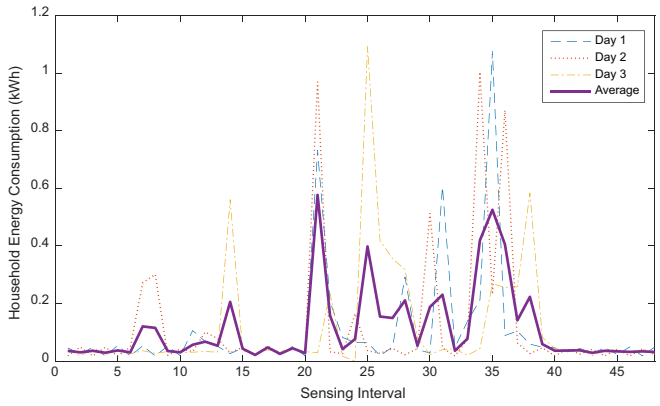


Fig. 3. Demonstration of energy consumption profiles of a typical Australian residential user [30].

selected retail plans. The rating values represent the users' preferences on the plans;

(3) For a target user, the PRS extracts his/her energy consumption features, and finds out the set of similar users of the target user. Then, the PRS applies a similarity metric to quantify the similarity degrees of the target user and users in the UKD;

(4) The PRS reads out the retail plan choices of the similar users, and uses an aggregation method to aggregate the choices of similar users. Based on the aggregation results, the PRS makes rating predictions of the target user on different retail plans, and sorts the plans based on the predicted ratings; and

(5) Finally, the top n electricity retail plans of the sorted list are recommended to the target user.

Implementation methodology of the key components of the proposed PRS are presented as follows.

A. Establishment of User Knowledge Database

As introduced before, the user knowledge database stores 2 kinds of user data:

Firstly, the household energy consumption data of users are directly collected by the AMI. Currently, the sampling frequency of the smart meter infrastructure ranges from minutes to hourly basis, depending on the hardware implementation. As a demonstration, Fig. 3 shows 3-day energy consumption profiles and averaged profile of a real residential user in New South Wales, Australia. Based on the energy consumption data, the users' key energy consumption features are extracted, shown in Table I.

Secondly, in the user knowledge database, each retail plan is associated with an identifier (ID), which is a positive integer. Then, for each user, its currently and historically selected retail plans in the last 2 years are represented by the plan IDs.

Combining above three kinds of information, the *user profile* of a user in UKD can be formed, represented in Table I.

B. User Similarity Calculation

In order to find out the similar users of a target user, a certain metric needs to be applied to calculate the similarity of different users. In conventional collaborative recommendation techniques, to decide whether recommend a target item to the target user or not, similarities between the target user and other users are calculated based on their ratings on the items that all users have rated [17]. This is based on an observation that "*if users have similar tastes on other items, they would have similar tastes on the target item*". For the electricity retail plan recommendation application in this paper, we transfer this observation to be "*if users have similar energy consumption patterns, they would have similar preferences and needs on the electricity retail plans*". And the concept of "commonly rated items" in the conventional recommender systems is mapped to the "shared energy consumption features" of the users, as listed in the "Energy Consumption Features" category of Table I.

TABLE I
USER PROFILE IN UKD

Category	Feature Name	Datatype
Energy Consumption Features	Averaged Monthly Consumption	Real
	Averaged Monthly Consumption in Summer and Winter Seasons	Real
	Averaged Monthly Consumption in Spring and Autumn Seasons	Real
	Averaged Weekly Consumption in Work Time	Real
	Averaged Weekly Consumption in Non-Work Time	Real
Retail Plan Information	IDs of selected retail plans in the last 2 years	Integer array

In conventional collaborative recommendation technique, similarities between the target user and each other user is calculated to get a numerical similarity degree. In the smart grid context, the number of end users could be very large, and the similarity calculation between the target user and all other users would thus lead to high computational costs. Therefore, in this paper we propose a strategy to reduce the computational cost of user similarity calculation. Firstly, users are clustered based on their energy consumption features, and then only users who are in the same cluster with the target user are considered as the similar user of the target user. The similarity metrics are then applied to calculate the similarity degrees between the target user and his/her similar users.

1) User Clustering: Firstly, users are clustered based on their energy consumption features. The ith user c_i is represented as a 5-dimensional vector $\mathbf{c}_i = [c_{i1}, c_{i2}, c_{i3}, c_{i4}, c_{i5}]$, where c_{ij} represents the jth energy consumption feature of the ith user shown in Table I.

Then, the fuzzy C-means (FCM) clustering method [31] is used, which is a proven efficient clustering method and has been applied in many industrial applications such as [32] and [33]. FCM uses a membership matrix to represent membership grads of the data points for multiple clusters. By iteratively updating cluster centers and membership grades for each data point, FCM moves the cluster centers to the right location within the dataset. More details about FCM can be found in [31].

2) User Similarity Calculation: The similarity degrees between the target user and other users are calculated. In the recommendation research, there are several available similarity calculation methods, such as the Pearson correlation, cosine-based approach, mean squared difference measure,

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etc. [17], [18]. In this paper, the cosine-based approach is applied, which calculates the similarity of two users (denoted as c_1 and c_2) as,

$$sim(c_1, c_2) = \cos(\overrightarrow{c_1}, \overrightarrow{c_2}) = \frac{\overrightarrow{c_1} \cdot \overrightarrow{c_2}}{\|\overrightarrow{c_1}\|_2 \times \|\overrightarrow{c_2}\|_2} = \frac{\sum_{i=1}^{D} c_{1i} c_{2i}}{\sqrt{\sum_{i=1}^{D} c_{1i}^2} \sqrt{\sum_{i=1}^{D} c_{2i}^2}} \quad (4)$$

where $sim(c_1, c_2)$ denotes the similarity degree of c_1 and c_2, which is a numerical value; $\overrightarrow{c_1}$ and $\overrightarrow{c_2}$ denote the feature vectors of the users c_1 and c_2; D is the dimension of the user feature vector; $\overrightarrow{c_1} \cdot \overrightarrow{c_2}$ denotes the dot-product between the users.

C. Aggregation of Similar Users' Choices

Choices of users on electricity retail plans need to be aggregately considered to generate the recommendation list of retail plans to the target user. In conventional recommender systems, the choices, preferences, and interests of the users are often reflected by their explicit ratings on the items. For example, a user can rate an electronic movie on the Web pages with the number of 'stars' or a non-negative integer to express his/her interest degrees on the movie. In some applications, it would be difficult to collect explicit ratings of the users, and some non-intrusive methods are therefore developed to implicitly infer the users' interests based on their behaviors. For example, a Web page recommender system infers the user's ratings on Web pages based on his/her click times [34].

In smart grid, it is often difficult to directly collect users' explicit ratings on the plans. Therefore, in this study we firstly propose a rating inference strategy to express users' interests, and then employ a weighted aggregation strategy to aggregate the similar users' ratings and predict the rating of the target user on a particular retail plan.

1) Rating Inference Strategy: In this study, we design a rating inference strategy to express the users' interests on retail plans and represent them as non-negative integers. We mainly consider retail plans selected by a user in the past 2 years, and use an integer in the rage of [0, 5] to represent a user's 'preferences on each plan. The larger value indicates the larger preference. The proposed rating inference strategy is as follows.

(a) For a given user, sort his/her historical used retail plans in last 2 years by time;
(b) Score the first retail plan (user's current plan) to be 5;
(c) Score the last two plans except for the current plan to be 3 and 2, respectively;
(d) Score all other historically selected plans to be 1;
(e) Score all the plans which have not been used by the user in the last two years to be 0.

Note that a same plan can be used multiple times by the user, and above rating inference strategy is simply applied in the individual plan choices. For example, if a user continuously uses a 6 month-contracted plan twice in the past year, then the rating of the user on that plan is $5 + 3 = 8$.

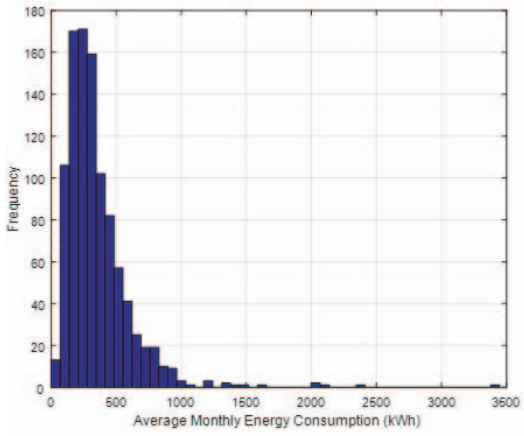


Fig. 4. Frequency distribution of the average monthly energy consumption of the 1,000 residential users.

2) Rating Aggregation: By applying above rating inference strategy, ratings of the users on retail plans can be obtained, which are then aggregated to generate the predicted rating of the target user on retail plans. In this study, we use the weighted sum function to do the aggregation (shown as Eq. (5)), which is the most commonly used aggregation function in PR [14].

$$r_{c,s} = k \sum_{c' \in \hat{C}} sim(c, c') \times r_{c',s} \quad (5)$$

where $r_{c,s}$ is the predicted rating of the target user c on retail plan s; k is the normalizing factor which is usually set as $k = 1/\sum_{c' \in \hat{C}} |sim(c, c')|$; $\hat{C}$ is the set of similar users of the target user c; $r_{c',s}$ is the rating of the similar user c' on plan s.

D. Recommendation

After calculating similarity degrees between the target user and other users and aggregating the users' preferences, predicted ratings of the target user on different retail plans are calculated. Finally, the retail plans are sorted with the descending order of the predicted ratings, and the top n plans are recommended to the target user.

V. Experiments

A. Simulation Data Preparation

1) Preparation of User Data and Retail Plans: We use 1-year smart meter data of 1,000 residential users in New South Wales, Australia, covering the period of Jan/2013 to Jan/2014. The data are available in [30]. The frequency distribution of the average monthly energy consumptions of the 1,000 users are shown in Fig. 4. For each user, we extract his/her energy consumption features listed in Table I. We then collect totally 46 electricity retail plans from the "Power-to-Choose" platform [4]. The plans are published by multiple retailers and each retailer is associated with a reputation level represented by the number of stars from 1 to 5. The plans are divided into 3 classes, shown in Table II (in Power-to-Choose platform, almost all the TOU plans are with variable rates).

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TABLE II
GENERAL INFORMATION OF COLLECTED 150 RETAIL PLANS

Class Index	Plan Characteristic	Number of Plans
C1	Fixed Rate, Non-TOU	20
C2	Variable Rate, Non-TOU	20
C3	Variable Rate, TOU	6



Fig. 5. User samples with randomly assigned plan class labels.

2) Electricity Retail Plan Selection Pattern Assumptions: Based on the collected residential energy consumption data and retail plans, in order to produce the data for simulation, we need to artificially associate the electricity plans to each of the 1,000 users. To simulate the practical situations, we consider following patterns:

Pattern 1 Users with large seasonal energy consumption deviations would prefer to choose the fixed rate plan; otherwise, the users would prefer the variable rate plan.

Pattern 2 Users with large energy consumption during the non-working periods would prefer the TOU plan.

Pattern 3 When the plan rates are close, users tend to choose the plans from the retailers with high reputations.

Pattern 4 When the plan rates are close, users tend to choose the plans with high renewable penetrations.

For a large group of users, above patterns are reasonable to a large extent. For example, the variable rate plans often have lower tariff rates than the fixed rate plans, but they are heavily affected by the power market. High energy consumptions in summer & winter seasons often imply the over high or over low ambient temperatures, which consequentially imply high network load and thus high risks of market price spikes. Also, pattern 4 tends to hold with the increase of the public's environmental awareness.

It is worth noting that the patterns assumed above are used for the simulation purpose. In real situations, there would be different trends or patterns existed in a particular group of users. However, as long as there exist certain patterns or trends for the choices of a group of users on electricity retail plans (in other words, as long as the assumption presented in Section IV holds), the proposed PRS would be able to extract the patterns from the collective behaviors of the users.

3) Associating Retail Plans With Users: Based on above patterns, for each user, we identify the class of plan which is mostly suitable for the user, and then assign the plan class label to the user. The user samples with the assigned plan class label are shown in Fig. 5. Then, we artificially and randomly associate the plans from the corresponding plan class to each user. In this simulation process, we ensure that the users have high probability to choose the plans with relatively low rates and come from retailers with relatively high reputation and renewable penetrations.

TABLE III
RECOMMENDATION PERFORMANCES OF THREE CROSS-VALIDATED CASES AND TWO BASE CASES

Case Index	$\overline{\lvert E \rvert}$	δ
Test Case 1	0.076	0.591
Test Case 2	0.077	0.515
Test Case 3	0.080	0.573
Base Case 1	0.12	0.87
Base Case 2	0.10	0.76

B. PRS Evaluation Settings

From the simulation dataset, 20% user profiles are randomly removed and used the *test dataset*. The remaining 80% user profiles are used as the *source dataset*. To evaluate the proposed PRS, we predict the rating of the target user in the test dataset on each retail plan, using only the data in the source dataset. 3-fold cross validation is performed to ensure the consistency of the rating prediction results.

Denote $\{r_1, r_2, \ldots, r_N\}$ and $\{p_1, p_2, \ldots, p_N\}$ as the real and predicted ratings of the test dataset, and denote $E = \{p_1 - r_1, p_2 - r_2, \ldots, p_N - r_N\}$ as errors. Then, the mean absolute error and error standard deviation are used to evaluate the rating prediction of the PRS, expressed as Eqs. (6) and (7):

$$\overline{|E|} = \frac{\sum_{i=1}^{N} |p_i - r_i|}{N} \tag{6}$$

$$\delta = \sqrt{\frac{\left(\sum \left(E - \overline{E}\right)^2\right)}{N}} \tag{7}$$

For comparison purpose, two base cases are designed.

(1) **Base case 1:** for each plan, the mean score received by the plan over all users in the source set is used as the predicted score for that plan. This base case represents an arbitrary/random recommendation, and can be used for evaluating the collaborative filtering mechanism in the proposed PRS.

(2) **Base case 2:** no user clustering is performed, and the user similarity is calculated between the target user and each other user. This case is set up for evaluating the user clustering mechanism in the proposed PRS.

The PRS is neither personalized nor accurate unless it has significantly lower values over both base case 1 (arbitrary recommendation) and base case 2 (without user clustering) on above two evaluation criteria.

C. Recommendation Results

We run both the base case and proposed PRS on the simulation dataset. We are only interest in the prediction errors on the samples with predicted ratings larger than 3, because only

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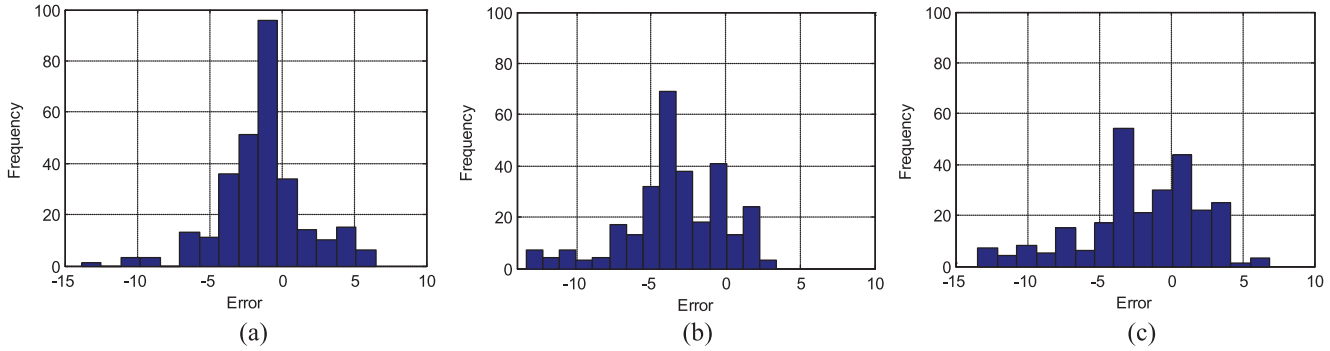


Fig. 6. Distribution of rating prediction error under the proposed PRS: test case 1 (a), base case 1 (b), and base case 2 (c).

TABLE IV
DEMONSTRATION OF TOP 5 PLANS SORTED BY POWER-TO-CHOOSE PLATFORM AND RECOMMENDED BY PRS

	Power-to-Choose Platform					Personalized Recommender System				
	Plan Index	Price/kWh	Retailer	Retailer Star	Renewable Rate	Plan Index	Price/kWh	Retailer	Retailer Star	Renewable Rate
1	p4	6.1 cent	Pennywise	2	5%	p8	6.6 cent	StarTex Power	4	10.1%
2	p5	6.1 cent	Pennywise	2	5%	p17	6.4 cent	Veteran	5	4%
3	p12	6.2 cent	Infinite Energy	2	5%	P18	6.7 cent	Veteran	5	4%
4	p17	6.4 cent	Veteran	5	4%	p22	7.2 cent	Reliant	4	100%
5	p8	6.6 cent	StarTex Power	4	10.1%	p4	6.1 cent	Pennywise	2	5%

Note: The results from Power-to-Choose are with the query settings: search date-30/06/2016; postcode-75015; estimated use-[500, 1000] kWh; plan type-fixed rate.

the plans with high ratings will be recommended to the user. Table III shows the values of $\overline{|E|}$ and δ under the two base cases and three cases generated in the 3-fold cross validation (denoted as test case 1, test case 2, and test case 3, respectively). The results show that the prediction performance of the three cases in the 3-fold cross validation are quite close, which proves the stability of the proposed recommendation framework. Table III also shows the values of $\overline{|E|}$ and δ under the 3 test cases are significantly smaller than those under the base case 1 (0.12 and 0.87, respectively) and base case 2 (0.10 and 0.76, respectively). As an illustration, we draw the distributions of errors under case 1 and the two base cases in Fig. 6. Comparing with base cases, it can be seen the error distribution under the PRS generally shows a desired bell shape, which indicates a good rating prediction performance.

Currently, there are some online ancillary tools developed to help users filter the electricity plans, such as the plan sorting function provided by Power-to-Choose platform [4] and the "Energy-Made-Easy" system [35]. These tools do not reveal the preference trends from the social information, but are purely price driven. User specifies plan conditions, and the ancillary tools sort the filtered plans with the ascending order of monthly costs. Since retail plan configurations (describeSection II) are complex for most of the residential users, many users would tend to choose the plans ranked on top of the price-based filtered list. This creates opportunities for some smart retailers to play a trick on their plan rakings and cheat the users. For example, there have been widely reported that some retailers play tricks on the plan configurations to create the "1-cent plans" to deliberately make their plans located on top of the price-based list, and the users actually pay much more electricity fees than what they expected [36], [37]. The social information filtering based PRS can help to filter such fraud plans for the users. In a large group of users, such plans could cheat some users at first, but will be gradually detected and abandoned by the users. The social information filtering based PRS would not recommend the user the fraud plans which are detected by the user groups.

As a demonstration, Table IV shows the top 5 filtered plans from the Power-to-Choose platform and top 5 recommended plans by the proposed PRS respectively, for a user with average month usage of 800 kWh. It can be seen that in the list filtered by the Power-to-Choose, the top 3 plans are filtered by retailers with only 2 stars, although these plans have lowest prices. In the list recommended by the PRS, 4 of the 5 filtered plans are from retailers with 4 or 5 stars and/or with high renewable penetrations, and also with relatively low prices. This is because in our simulation settings, these plans are more preferred by the similar users of the target user.

Lastly, it is worth noting that above discussions do not mean that the existing ancillary and the proposed PRS are conflict systems. They just use different filtering mechanisms and can coordinately work together to provide information to the users, just as what have been adopted by most of the current e-commerce websites (e.g., Amazon, E-bay, etc.).

VI. CONCLUSION

In this paper, a PRS is developed for the smart grid residential end users. The proposed PRS is based on the collaborative recommendation technique, and filter the retail plans for the target user by extracting the preferences of the similar users on the plans. For a target user, his/her similar users on the aspects of energy consumption features are firstly found out by the PRS, and then the ratings of the target users on different plans

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are predicted by aggregating the ratings of the similar users. Experiments are conducted to validate the proposed PRS.

One limitation of the CF method is the cold-start problem. A new plan which is not rated by other users will be never recommended to the target user. In future, the improvement strategies can be incorporated to relieve the cold-start problem. The authors are also working on developing other kinds of smart grid personalized recommender systems.

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REFERENCES

[1] J. Giri, D. Sun, and R. Avila-Rosales, "A more intelligent grid," *IEEE Power Energy Mag.*, vol. 7, no. 2, pp. 34–40, Mar. 2009.

[2] F. Luo *et al.*, "Enabling the big data analysis in the smart grid," in *Proc. IEEE PES Gener. Meeting*, Denver, CO, USA, 2015, pp. 1–5.

[3] F. Luo *et al.*, "Cloud-based information infrastructure for next-generation power grid: Conception, architecture, and applications," *IEEE Trans. Smart Grid*, vol. 7, no. 4, pp. 1896–1912, Jul. 2015.

[4] *Power-to-Choose*. Accessed: Jun. 2017. [Online]. Available: http://www.powertochoose.org/

[5] B. Ramanathan and V. Vittal, "A framework for evaluation of advanced direct load control with minimum disruption," *IEEE Trans. Power Syst.*, vol. 23, no. 4, pp. 1681–1688, Nov. 2008.

[6] F. Luo *et al.*, "Optimal dispatch of air conditioner loads in southern China region by direct load control," *IEEE Trans. Smart Grid*, vol. 7, no. 1, pp. 439–450, Jan. 2015.

[7] F. Luo *et al.*, "An operational planning framework for large-scale thermostatically controlled load dispatch," *IEEE Trans. Ind. Informat.*, vol. 13, no. 1, pp. 217–227, Feb. 2017.

[8] C. Vivekananthan, Y. Mishra, G. Ledwich, and F. Li, "Demand response for residential appliances via customer reward scheme," *IEEE Trans. Smart Grid*, vol. 5, no. 2, pp. 809–820, Mar. 2014.

[9] *SmartAC Program*. Accessed: Jun. 2017. [Online]. Available: http://www.pge.com/en/myhome/saveenergymoney/plans/smartac/index.page?WT.mc_id=Vanity_smartac

[10] M. H. Albadi and E. F. EI-Saadany, "A summary of demand response in electricity markets," *Electric Power Syst. Res.*, vol. 78, no. 11, pp. 1989–1996, 2008.

[11] D. W. Caves and L. R. Christensen, "Econometric analysis of residential time-of-use electricity pricing experiments," *J. Econometrics*, vol. 14, no. 3, pp. 287–306, 1980.

[12] J. Y. Choi, S.-H. Rim, and J.-K. Park, "Optimal real time pricing of real and reactive powers," *IEEE Trans. Power Syst.*, vol. 13, no. 4, pp. 1226–1231, Nov. 1998.

[13] K. Herter, "Residential implementation of critical-peak pricing of electricity," *Energy Policy*, vol. 35, no. 4, pp. 2121–2130, 2007.

[14] M. A. Piette, D. S. Watson, N. Motegi, S. Kiliccote, and E. Linkugel, "Automated demand response strategies and commissioning commercial building controls," in *Proc. 14th Nat. Conf. Build. Commissioning*, San Francisco, CA, USA, Apr. 2006, pp. 1–13.

[15] F. Luo, G. Ranzi, X. Wang, and Z. Y. Dong, "Service recommendation in smart grid: Vision, technologies, and applications," in *Proc. 9th Int. Conf. Service Sci.*, 2016, pp. 1–8.

[16] F. Luo *et al.*, "Non-intrusive energy saving appliance recommender system for smart grid residential users," *IET Gener. Transm. Distrib.*, vol. 11, no. 7, pp. 1786–1793, Nov. 2017.

[17] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions," *IEEE Trans. Knowl. Data Eng.*, vol. 17, no. 6, pp. 734–749, Jun. 2005.

[18] J. Bobadilla, F. Ortega, A. Hernando, and A. Gutiérrez, "Recommender systems survey," *Knowl. Based Syst.*, vol. 46, pp. 109–132, Jul. 2013.

[19] *Reforming Power Markets in Developing Countries*. Accessed: Jun. 2017. [Online]. Available: http://siteresources.worldbank.org/INTENERGY/Resources/Energy19.pdf

[20] *Roadmap of Retail Electricity Market Reform in China: Assisting in Mitigating Wind Energy Curtailment*. [Online]. Available: http://iopscience.iop.org/article/10.1088/1742-6596/52/1/012031/pdf

[21] *Was Amazon's Recommendation Engine Crucial to the Company's Success?* Accessed: Jun. 2017. [Online]. Available: https://www.quora.com/Was-Amazon-s-recommendation-engine-crucial-to-the-company-s-success

[22] X. Wang, F. Luo, Y. Qian, and G. Ranzi, "A personalized electronic movie recommendation system based on support vector machine and improved particle swarm optimization," *PLoS ONE*, vol. 11, no. 11, 2016, Art. no. e0165868.

[23] S. Li *et al.*, "A new QoS-aware Web service recommendation system based on contextual feature recognition at server-side," *IEEE Trans. Netw. Service Manag.*, vol. 14, no. 2, pp. 332–342, Jun. 2017.

[24] W. Hill, L. Stead, M. Rosenstein, and G. Furnas, "Recommending and evaluating choices in a virtual community of use," in *Proc. Conf. Human Factors Comput. Syst.*, Denver, CO, USA, 1995, pp. 194–201.

[25] M. Balabanović and Y. Shoham, "Fab: Content-based, collaborative recommendation," *Commun. ACM*, vol. 40, no. 3, pp. 66–72, 1997.

[26] J. S. Breese, D. Heckerman, and C. Kadie, "Empirical analysis of predictive algorithms for collaborative filtering," in *Proc. 14th Conf. Uncertainty Artif. Intell.*, Madison, WI, USA, Jul. 1998, pp. 43–52.

[27] J. Delgado and N. Ishii, "Memory-based weighted-majority prediction for recommender systems," in *Proc. ACM SIGIR Workshop Recommender Syst. Algorithms Eval.*, 1999, pp. 1–10.

[28] Y. Chien and E. George, "A Bayesian model for collaborative filtering," in *Proc. 7th Int. Workshop Artif. Intell. Stat.*, Fort Lauderdale, FL, USA, 1999, pp. 1–6.

[29] D. Y. Pavlov and D. M. Pennock, "A maximum entropy approach to collaborative filtering in dynamic, sparse, high-dimensional domains," in *Proc. 15th Annu. Conf. Neural Inf. Process. Syst.*, Vancouver, BC, Canada, 2002, pp. 1465–1472.

[30] *Smart Grid, Smart City*. Accessed: Jun. 2017. [Online]. Available: http://www.industry.gov.au/ENERGY/PROGRAMMES/SMARTGRIDSMARTCITY/Pages/default.aspx

[31] J. Bezdek, R. Ehrlich, and W. Full, "FCM: The fuzzy *c*-means clustering algorithm," *Comput. Geosci.*, vol. 10, nos. 2–3, pp. 191–203, 1984.

[32] Y. Chen *et al.*, "Collector system layout optimization framework for large-scale offshore wind farms," *IEEE Trans. Sustain. Energy*, vol. 7, no. 4, pp. 1398–1407, Oct. 2016.

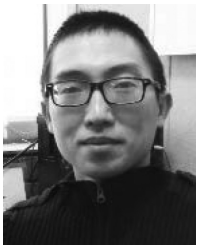
[33] Y. Chen *et al.*, "A novel technique for the optimal design of offshore wind farm electrical layout," *J. Modern Power Syst. Clean Energy*, vol. 1, no. 3, pp. 258–263, 2013.

[34] D. Chakrabarti, D. Agarwal, and V. Josifovski, "Contextual advertising by combining relevance with click feedback," in *Proc. ACM Int. World Wide Web Conf.*, Beijing, China, 2008, pp. 417–426.

[35] *Energy Made Easy.* Accessed: Jun. 2017. [Online]. Available: https://www.energymadeeasy.gov.au/

[36] *Misleading Rates of Power-to-Choose Website*. Accessed: Jun. 2017. [Online]. Available: http://fuelfix.com/blog/2016/06/29/power-to-choose-website-tweaked-to-weed-out-misleading-rates/

[37] *The 5 Problems With Power-to-Choose.* Accessed: Jun. 2017. [Online]. Available: https://www.searchforenergy.com/blog/powertochoose/



Fengji Luo (M'13) received the B.S. and M.S. degrees in software engineering from Chongqing University, China, in 2006 and 2009, respectively, and the Ph.D. degree in electrical engineering from the University of Newcastle, Australia, in 2013. He is currently a Post-Doctoral Research Associate with the School of Civil Engineering, University of Sydney, Australia. His research interests include the demand side management, active distribution networks, renewable energy, and computational intelligence and its applications in smart grid. He has published over 90 papers in above areas.

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Gianluca Ranzi (M'16) received the degree in management and production engineering with the Politecnico di Milano, Italy, the B.E. (First Class Hons.) degree with the University of Wollongong, Australia, the degree in civil engineering with the Universita' Politecnica delle Marche, Italy, and the Ph.D. degree with the University of New South Wales, Australia. He is currently an ARC Future Fellow, a Professor, and the Director of the Centre for Advanced Structural Engineering, University of Sydney, Australia. His research interests include high performance building, building-to-grid technology, and demand side management.



Xibin Wang received the M.S. degree in computer science from Guizhou University, China, in 2012 and the Ph.D. degree in computer science from Chongqing University, China, in 2015. He is currently a Lecturer with the School of Electrical and Information Engineering, Guizhou Institute of Technology, China. His research interests include computational intelligence, data mining and business intelligence, and machine learning.



Zhao Yang Dong (M'96–SM'04–F'07) received the Ph.D. degree from the University of Sydney, Australia, in 1999. He is currently with the University of NSW, Sydney, Australia. His immediate role was a Professor and the Head of the School of Electrical and Information Engineering, University of Sydney, Australia. He was an Ausgrid Chair and the Director of the Centre for Intelligent Electricity Networks, University of Newcastle, Australia. He also worked as a Manager (system planning) with Transcend Networks (now TASNetworks), Tasmania, Australia. His research interests include power system planning, power system stability, smart grid, renewable energy systems, computational intelligence, and data mining. He is an Editor of the IEEE TRANSACTIONS ON SMART GRID, the IEEE TRANSACTIONS OF SUSTAINABLE ENERGY, the IEEE Power Engineering Letters, and *IET Renewable Power Generation*.

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